

Research on the technical scheme of multi-stack common rail fuel cell engine based on the demand of commercial vehicle

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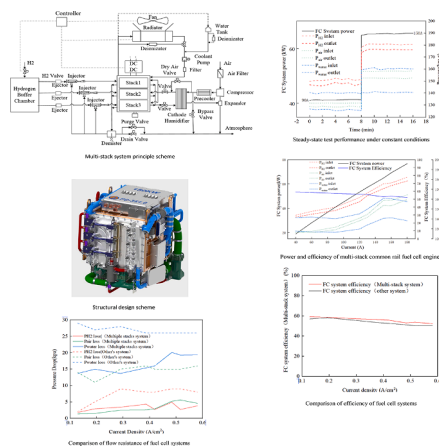
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HIGHLIGHTS

- A scheme of multi-stack common rail fuel cell system with high performance is proposed.
- The design, trial production and test verification of multi-stack common rail fuel cell engine have been completed.
- The intelligent control design and test verification based on the independent hydrogen injection of single stack and the mixing of back hydrogen in multiple stacks are completed.
- The advantages of multi-stack common rail system scheme, such as high efficiency, good uniformity and consistency, have been verified through experiments.

GRAPHICAL ABSTRACT



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ABSTRACT

At present, most fuel cell engines are single-stack systems, and high-power single-stack systems have bottlenecks in meeting the power requirements of heavy-duty trucks, mainly because the increase in the single active area and the excessive number of cells will lead to poor distribution uniformity of water, gas and heat in the stack, which will cause local attenuation and reduce the performance of the stack. This paper introduces the design concept of internal combustion engine, takes three-stack fuel cell engine as an example, designs multi-stack fuel cell system scheme and serialized high-voltage scheme. Through Intelligent control technology of independent hydrogen injection based on multi-stack coupling, the hydrogen injection inflow of each stack is controlled online according to the real-time anode pressure to achieve accurate fuel injection of a single stack and ensure the consistency between multiple stacks. proves the performance advantage of multi-stack fuel cell engine through theoretical design, intelligent control and test verification, and focuses on analyzing the key technical problems that may exist in multi-stack consistency. The research results provide a reference for the design of

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multi-stack fuel cell engines, and have important reference value for the powertrain design of long-distance heavy-duty and high-power fuel cell trucks.

1. Introduction

In the context of carbon neutrality, Chinese energy field is facing key opportunities and challenges, the development of clean energy has become an inevitable trend, the global commercial heavy truck in the market promotion process will be subject to a variety of use conditions [1], especially long-distance heavy-duty freight vehicles, in the driving range, charging time and other aspects of the use of certain problems, such as short mileage, long charging time [2]. There are also certain technical problems in the power battery, such as large battery capacity, high cost, and heavy chassis [3]. Through the analysis of the economy of long-distance heavy-haul transportation, its economic performance weakens with the increase of load weight and mileage [4]. Therefore, in the field of heavy-duty long-distance freight transport, it is necessary to seek other power configuration technology routes to enhance the market penetration capacity of new energy commercial vehicles [5].

Hydrogen energy is recognized as a clean energy source, and accelerating the development of hydrogen energy can promote the important mission of achieving carbon neutrality [6]. In the development of hydrogen energy, hydrogen fuel cell vehicles are an important breakthrough to promote the application of hydrogen energy, and China, Japan and South Korea, Europe and the United States have increased investment in the research and development of hydrogen fuel cell vehicles [7]. According to the different objects used, cars can be divided into passenger cars and commercial vehicles. Due to the high price cost of fuel cell passenger cars and infrastructure restrictions such as hydrogen refueling stations, the promotion of passenger cars has been greatly hindered [8]. While high-power commercial vehicles run on a relatively fixed route, the dependence on the breadth and number of hydrogen refueling stations is low, and for long distance driving and high power performance requirements, commercial fuel cell vehicles have more advantages. Therefore, in the initial development stage of fuel cell technology, Chinese research and development goal focuses on high-power fuel cell commercial vehicles [9].

For high-power fuel cell engines, there are currently more single-stack systems. Generally 49 tons of heavy truck powertrain demand power of about 380 kW, common such as Toyota Alpha and Hyundai Xcient as the representative of hydrogen heavy commercial vehicles, fuel cell engine power is about 200 kW, if the single stack become to 200 kW power or even above, It is necessary to increase the active area of a single cell, or increase the number of cells [10], which will lead to poor uniformity of water, gas and heat distribution, large flow resistance, unequal current density distribution problems, and the increase in the number of cells will reduce the vibration reliability [11].

Zhang Caizhi et al proposed that the Multi-stack fuel cell system can greatly reduce cost and efficiency loss, he claimed that comparing with single-stack fuel cell system, multi-stack fuel cell system has the advantages of low replaceable cost, excellent redundancy, higher efficiency and longer life. And, low durability, limited output power, no-output in case of fault and poor redundancy are increasingly visible drawbacks to single-stack fuel cell system in applications. He use two sub-health fuel cell stacks with poor consistency to study the dynamic and transient characteristics, power adaptive allocation and output stability, The result is the fuel cells run steadily and less than 50 ms transient process in both structures [12]. He also proposed an improved overall efficiency maximization strategy (I-OEMS) that combines a predictive soft-loading method to improve the load state of sub-stacks while ensuring the approximate maximum efficiency. The simulation results show that the strategy can effectively limit the loading speed, reduce the start/stop frequency, and operate the auxiliary sub-stack away from idling/open-circuit, so as to enhance the durability [13].

However, there is no DC/DC device in the fuel cell system, it is quite difficult to prevent current backflow which may leading the stack to reverse polarity. Moreover, to ensure the output voltage of the branch to be consistent with the bus bar, the output voltage range of the fuel cell will be narrow and relatively unstable, and it is difficult to meet the power demand range of the vehicle.

Therefore, this paper introduces the concept of multi-stack common rail fuel cell engine, solves the problem of high-power development of fuel cell engine for commercial vehicles, realizes the serialization of fuel cell engine products, the standardization of single stack, and the modularization of the whole stack. Starting from the demand of 49 tons long-distance heavy truck, the powertrain is top-down designed. The fuel cell engine power coverage range from 100 kW to 380 kW, while solving key technical problems such as multi-stack consistency etc..

For multi-stack system, the control of the fuel supply of the hydrogen subsystem is the key to keeping the output of the fuel cell system stable and durable. Bo Zhang, et al proposed a new structure of hydrogen circulation pump which can reducing the parasitic power by using the pressure potential energy of hydrogen flow. Through modeling simulation and control strategy development, the parasitic power consumption of the circulating pump is effectively reduced about two-thirds under the fuel cell vehicle operation condition [14]. In this paper, each stack of the fuel cell system is matched with a separate hydrogen injection valve and ejector. Through Intelligent control technology of independent hydrogen injection based on multi-stack coupling, the hydrogen injection inflow of each stack is controlled online according to the real-time anode pressure to achieve accurate fuel injection of a single stack and ensure the consistency between multiple stacks.

At the same time, this article also referred to hybrid power supply systems and intelligent energy managements for unmanned aerial vehicles in public services which Zhang Caizhi et al proposed [15], and a novel self-learning energy management strategy (EMS) for fuel cell hybrid electric vehicles (FCHEV) which Liang Guo et al proposed [16]. Synthesize fuzzy logic, minimum principle, data-driven intelligent control and other artificial intelligence energy management strategies, the system can learn itself through interactions with environment to achieve multi-objective energy management optimization control.

2. Commercial vehicle demand analysis

Take 49 tons heavy truck as an example, while taking into account the power requirements of other heavy vehicles, quasi-heavy vehicles, medium vehicles, as shown in Table 1.

2.1. Statistical analysis of commercial vehicle working conditions

According to the national standard "Heavy Commercial Vehicle Fuel Consumption Measurement Method", the 49 ton tractor is suitable for the national standard test condition CHTC TT condition of the semi-tractor, the maximum speed is 88 km/h, the average speed is 46.42 km/h, the maximum acceleration is 1.19 m/s², and the maximum braking deceleration is -1.22 m/s², as shown in Fig. 1.

Table 1
Power requirements of commercial vehicles.

Serial number	1	2	3	4	5
Representative model	Middle-sized (12 ton)	Quasi-heavy(14 ton)	Quasi-heavy(16 ton)	6 × 2 Heavy	6 × 4 Heavy(49 ton)
Power demand (kW)	100	150	200	300	350

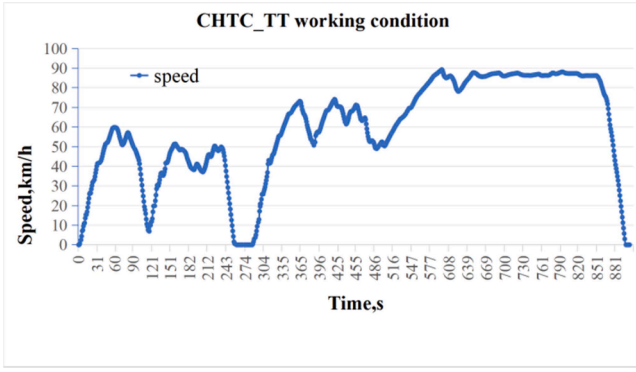


Fig. 1. CHTC_TT working condition the 49 ton tractor used.

According to the analysis of the power distribution of the vehicle power demand under the working conditions, the power segment below 20 kW accounts for 1.11 %, the 80–200 kW power segment accounts for 41.51 %, accounting for a relatively large proportion, the 200–280 kW power segment accounts for 19.69 %, and the 280–400 kW power segment accounts for 26.92 %, as shown in Fig. 2.

2.2. Dynamical system configuration analysis

Based on the technical benchmark, scheme research and simulation results of various technical routes of international commercial vehicles, the dynamic system configuration research is carried out. The main research content is divided into three aspects:

- (1). The research on the complete vehicle configuration of commercial vehicles. Based on the information such as user demand, regulatory requirements and market competitiveness evaluation, the research comprehensively evaluates three possible vehicle configuration schemes (including extended range, hybrid and full-power fuel cell) from the aspects of powertrain characteristics, assembly characteristics and other functions, as shown in Fig. 3. The three powertrain configurations adopt the same vehicle and transmission system parameters. The vehicle has a full load mass of 49 tons, a windward area of 7.434 m², a tire dynamic radius of 507 mm, double motors peak power of 2 × 125 kW, and peak torque of 2 × 450 Nm. However, the parameter matching between fuel cell systems and power batteries is not the same. The commonly used power range of fuel cells with extended range configuration is 55–100 kW, and the backup power range is 100–125 kW; The power battery has a capacity of 2 × 170 Ah and a power of 204 kWh. The commonly used power

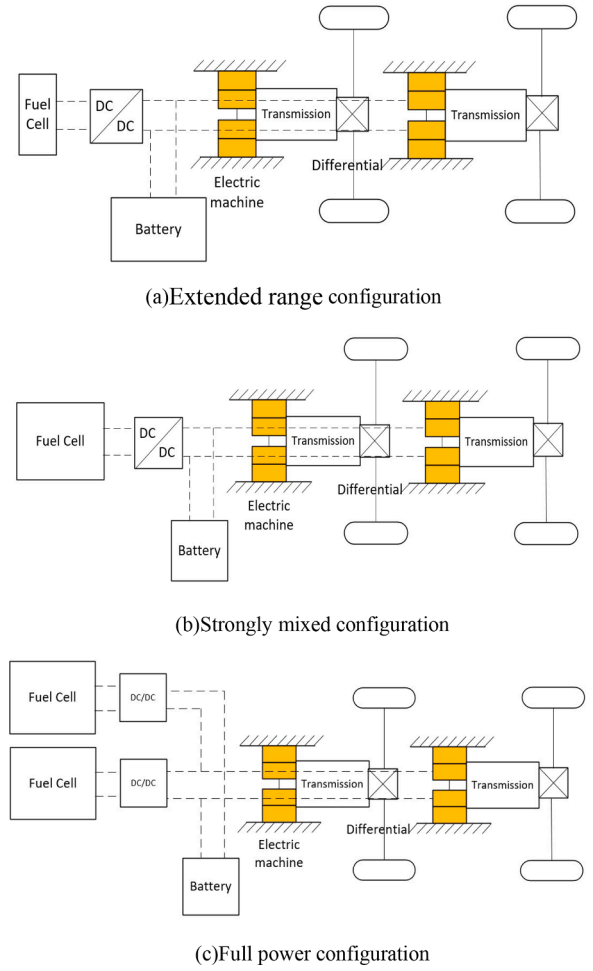


Fig. 3. Three possible dynamic system configuration scheme (including extended range, hybrid and full-power fuel cell).

range for fuel cells with strong mixing configuration is 110–200 kW, and the backup power range is 200–255 kW; The power battery has a capacity of 2 × 70 Ah and a power of 84 kWh. The commonly used power range for fuel cells with full power configuration is 160–300 kW, and the backup power range is 300–380 kW; The power battery has a capacity of 2 × 50 Ah and a capacity of 60 kWh.

- (2). Modeling and parameter matching of commercial vehicle and powertrain, based on vehicle development performance

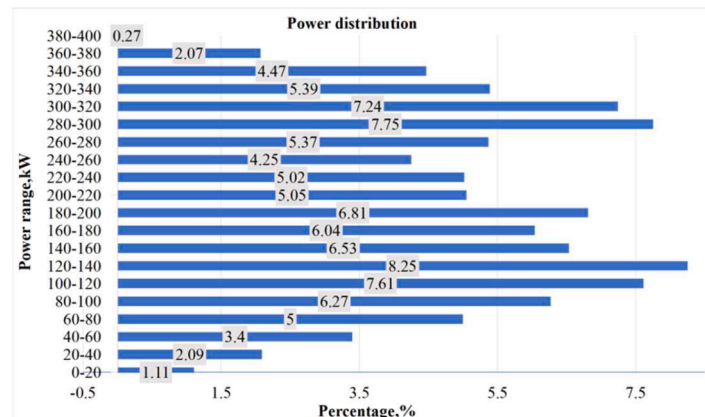


Fig. 2. Percentage distribution of power required by the vehicle powertrain (by time).

indicators, building vehicle model, powertrain (engine, fuel cell stack, key accessories, motor, battery, transmission, reducer) model, control algorithm model, energy management model, working condition model, etc. based on Cruise and Simulink. Construct vehicle powertrain simulation platform. With the minimum fuel (or hydrogen or power consumption) economy as the goal, power performance as the constraint condition of power matching and efficiency matching combined with powertrain parameter matching, so that the engine or fuel cell, motor work in their own efficient zone, complete the main performance and main parameter design of the powertrain. The dynamic and economic indexes of different working conditions are analyzed.

- (3). Energy management and optimization of commercial vehicle and powertrain systems, analyzing different control strategies, such as fuzzy logic, minimum principle, data-driven intelligent control and other energy management strategies, while considering the coupling relationship between fuel cell stack durability, power battery durability and economy, to achieve multi-objective energy management optimization control.

2.3. Dynamic system simulation results

Based on the model simulation analysis, it is concluded in Table 2 below that although the average efficiency of the full-power configuration is higher, the fuel cell acquisition cost and total ownership cost are higher.

The extender configuration basically works at the highest power point, the average efficiency is low, only 43–47 %, and the power battery is large, the vehicle weight increases greatly, resulting in an increase in the total cost of ownership.

The strong hybrid configuration has better power distribution and average efficiency, so it is preferred to choose the strong hybrid configuration. If it can be plugged in, it can further improve the economy, and the total cost of ownership and profit are the best.

To sum up, the 49-ton long-distance tractor is selected to select a strong hybrid configuration scheme, and the rule-based energy management strategy is designed as follows:

- (1). Fuel cell engine starting conditions: battery's SOC value less than 90 %.
- (2). Fuel cell engine shutdown conditions: car parking or battery's SOC value greater than 85 %.
- (3). Fuel cell engine operation rules:
 - 1) When the battery's SOC is greater than 90 %, the fuel cell stops until the battery's SOC is less than 85 %, and the fuel cell starts.
 - 2) When the battery's SOC is less than 85 %, and the fuel cell output power is guaranteed to be in the common power range, the output power is the penalty factor A (SOC) *Preq, and exit

when the battery's SOC is less than 30 % or battery's SOC > 90 %.

- 3) When battery's SOC is less than 30 %, the fuel cell outputs 255 kW until battery's SOC value is greater than 35 %.

Based on the above vehicle energy management strategy, the required output power distribution of the fuel cell engine is obtained through the calculation and decomposition of the powertrain, as shown in Fig. 4 below.

3. Multi-stack common rail fuel cell engine system scheme

3.1. Fuel cell engine product series design

According to the 49 tons of fuel cell heavy truck and taking into account the needs of other vehicle models, it is determined that the fuel cell engine power changes at every 50 kW interval, so the corresponding system power of a single stack is 50 kW. In this way, the fuel cell engine system with the same standardized stack in series or combined to synthesize multiple stack groups can cover the power needs of a variety of models. In addition, through the simulation analysis of three powertrain configurations of the 49 ton tractor and the power demand of the fuel cell engine under CHTC_TT condition, it is concluded that different powertrain configurations should be set up different common working areas and backup working areas of the fuel cell engine, so the product series of multi-stack common rail fuel cell engine is designed in Table 3.

3.2. Principle design of multi-stack common rail fuel cell engine system

The multi-stack common rail fuel cell system consists of the stacks, the air supply subsystem, the hydrogen supply subsystem, the thermal management subsystem and the control subsystem. The principle design is carried out for the demand boundary of a 49-ton heavy truck in China, as shown in Fig. 5. The hydrogen subsystem uses the combination of common rail injection system and ejector, and each stack can spray and cycle hydrogen independently to adjust the consistency between each stack. The air subsystem uses a separate air compressor design with energy recovery, which distributes air evenly to the individual stacks through pipelines. The thermal management subsystem uses a single high-voltage shielded water pump, and multiple stacks of cooling waterways are designed in parallel. The control subsystem controls the coordination of the three subsystems of water, hydrogen and air to ensure the efficient and stable operation of the whole system [17].

Single stack fuel cell engine system generally uses a single DC/DC voltage conversion to meet the requirements of the vehicle voltage platform. Since the voltage platform of commercial vehicle is 600 V, according to the structure and characteristics of multi-stack fuel cell engines, the high-voltage schemes of 2-stack, 3-stack, 4-stack and 6-

Table 2
Simulation result data of dynamic system.

Index	Extended range configuration	Strongly mixed configuration	Full power configuration
Hydrogen consumption, kg/100 km	11.64	10.52	10.12
Power consumption, kWh/100 km	0.9023	−0.2	0.2803
Average efficiency, %	47.04 %	52.99 %	54.85 %
FCS common power requirements, kW	55–100	110–200	160–300
FCS backup power requirements, kW	100–125	200–255	300–380
Power battery power requirements, kW	139.1	72.1	51.52

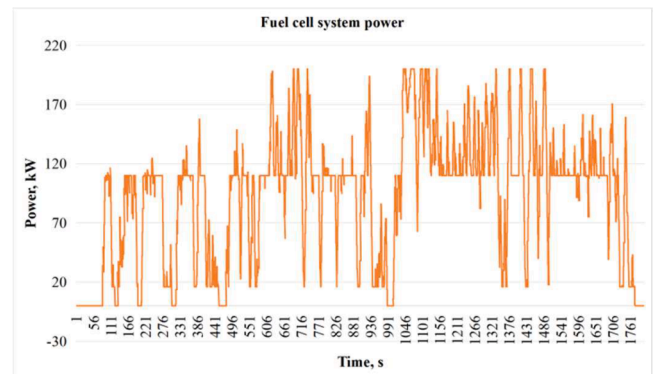


Fig. 4. Required output power distribution of fuel cell engine under CHTC_TT condition.

Table 3

Multi-stack common rail fuel cell engine product series.

Performance	S2-FCS	S3-FCS	S4-FCS	S6-FCS
Idle power, kW	8	12	16	24
Normal power area, kW	55–100	80–150	110–200	160–300
Reserve power area, kW	100–125	150–190	200–255	300–380
Loading speed, kW/s	>20	>20	>20	>20
Unloading speed, kW/s	>20	>20	>20	>20
Peak efficiency, %	60	60	60	60
Maximum efficiency of the normal area, %	57	57	57	57
Maximum efficiency of the Reserve area, %	51	51	51	51
Normal temperature start-up, s	35	35	35	35
−30°C Start, s	120 s	120 s	120 s	120 s
Normal temperature shutdown, s	<50	<50	<50	<50
Cold shutdown, s	<180	<180	<180	<180
Level of protection	IP67	IP67	IP67	IP67
Maximum noise, dBA	75	80	85	90
Hydrogen emission concentration	<4 %	<4 %	<4 %	<4 %

stack fuel cell systems are designed, respectively, as shown in Fig. 6. The high-voltage scheme of the two-stack system uses two stacks in series, and the boost DC/DC input voltage is 309–402 V. The high-voltage scheme of the three-stack system adopts three stacks in series, and the step-up DC/DC input voltage is 464–630 V. There are two high-voltage schemes for the four-stack system. One uses two stack in series and then in parallel, and the booster DC/DC input voltage is 309–402 V; the other uses four stacks in series, and the booster DC/DC input voltage is 618–840 V. There are two high-voltage schemes for the six-stack system. One uses three stacks in series and then parallel, and the boost DC/DC input voltage is 464–630 V; the other uses three stacks in series, and each matches the boost DC/DC, and the input voltage is 464–630 V.

3.3. Intelligent control of independent hydrogen injection based on multi-stack coupling

Since the anode control target is multiple stack inlet pressure and inlet flow, and the actuator is the duty cycle of hydrogen injection valve and the open and close control of hydrogen purge and drainage valve, the anode determines whether the current hydrogen supply is suitable through pressure control and flow estimation because the existing hydrogen flow meter technology cannot achieve on-board applications. In order to improve the efficiency of the system, each stack hydrogen circuit uses an intelligent and independent injection control scheme.

The hydrogen purge valve and drainage valve adopt the intelligent

control scheme based on model prediction. The control reference value is obtained by estimating the water production and nitriding amount according to the current discharge current, and then real-time correction is made according to the current average voltage and minimum voltage of the stack. When a single cell voltage is too low, the opening time of the hydrogen purge valve and drainage valve are intelligently controlled. The opening of the hydrogen purge valve and drainage valve will have a certain impact on the inlet pressure and flow rate of the stack. Therefore, the change of the opening degree of the hydrogen injection valve is followed by online control to ensure the stable hydrogen supply.

The theoretical hydrogen consumption of a single hydrogen injection valve is obtained from the corresponding current of the stack, and the hydrogen excess ratio is determined by bench calibration test, and then a flow control target value is obtained. Then, according to the anode outlet pressure, the pressure behind the gas-water separator and the specifications of the hydrogen purge valve, the gas flow rate is calculated when the hydrogen purge valve is opened. Finally, when the hydrogen purge valve or drain valve is opened, the average consumption flow rate is added to the target flow value of the hydrogen injection valve to obtain the final flow control value.

The control value of single anode inlet pressure is determined by the target value of anode inlet pressure and PID adjustment value, and the target value of anode inlet pressure is intelligently calculated according to the target value of cathode inlet pressure and the target pressure difference of anode and cathode. PID control takes the difference between the target value and the actual value of the anode into the stack as the feedback input, and obtains the adjustment value according to the proportion, integral and differential. The upper limit and lower limit of the adjustment are set for the product component, and the product component is reset according to the reset signal to perform supersaturation control.

4. Multi-stack common rail fuel cell engine component design and layout design

4.1. Stack design

According to the definition of the product series, the common working area of the fuel cell engine is 80–150 kW, in order to reduce the hydrogen consumption of the vehicle in the common speed zone of 100 km, it is necessary to improve the efficiency of the fuel cell engine in the 80–150 kW range, according to the power output characteristics of the stack, while taking into account the efficiency and life and other comprehensive factors. The common working area of the stack is defined as 0.6–1.2 A/cm², so that the stack often works in the ohmic polarization

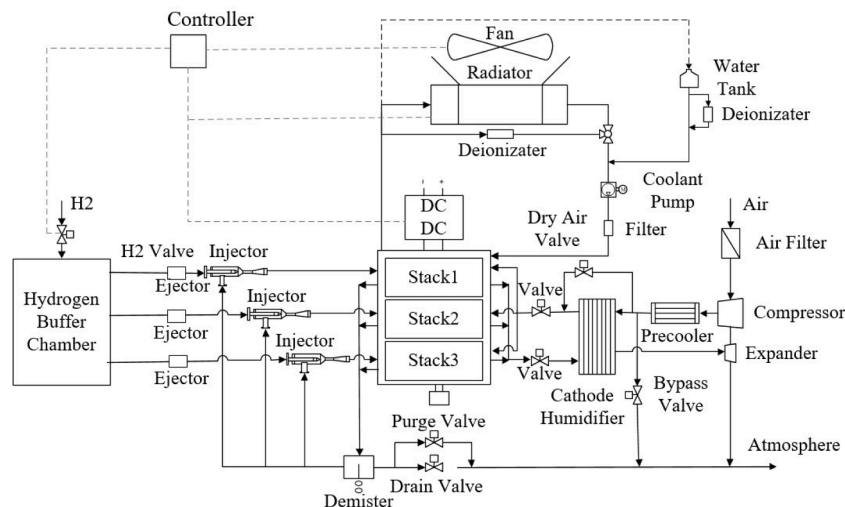


Fig. 5. Schematic diagram of multi-stack common-rail fuel cell system (with three-stack examples).

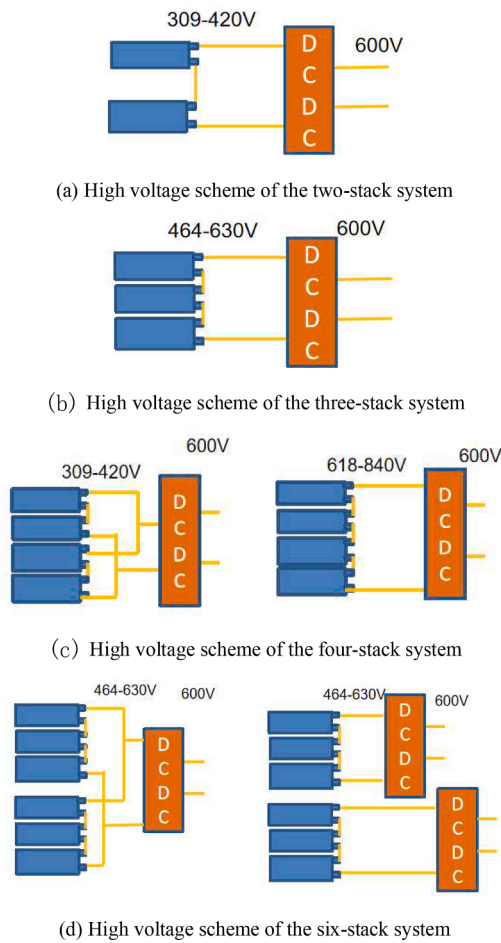


Fig. 6. High voltage scheme of the multi-stack system.

region.

Based on the previous simulation results of the powertrain configurations, the power demand of the fuel cell engine is decomposed. Taking the typical and simple three-stack system as an example, the active area of the stack is 310 cm^2 and the number of cells is 210 according to the forward design of the vehicle voltage platform and layout space, so that the stack can work at the idle point and the common working area of $0.6\text{--}1.2 \text{ A/cm}^2$. Single cell voltage range is $0.8\text{--}0.73 \text{ V}$. The performance parameters of the stack are shown in Table 4.

4.2. Parts design

In the fuel cell engine, the energy is mainly provided by the stack, and other components as external accessories must meet the use conditions of the stack, including the inlet pressure, flow rate and meet the heat dissipation requirements of the stack, so the main parameters of the

Table 4
Performance parameters of fuel cell stack.

Key parameter	Idle point	Common point 1	Common point 2
Current density	0.1 A/cm^2	0.6 A/cm^2	1.2 A/cm^2
Cell voltage	0.87 V	0.8 V	0.736 V
Total cell number	630	630	630
Current	31 A	186 A	372 A
Stack power	17 kW	92 kW	172.5 kW
Air excess coefficient	4	2	1.8
Air inlet pressure	140 kPa	190 kPa	220 kPa
Hydrogen excess coefficient	3	1.6	1.5
Hydrogen inlet pressure	150 kPa	200 kPa	230 kPa

three circuits of hydrogen, air and cooling are mainly calculated according to the needs of the stack operating conditions for the matching and selection of components of each subsystem. In addition to considering the use conditions of the stack (maximum temperature, pressure, flow, etc.), it is also necessary to consider the environmental limit temperature (to meet the use needs of cold or extremely hot areas), high altitude (to meet the use needs of plateau areas), etc., to determine the requirements of parts.

The air compressor chooses the energy recovery scheme with the expansion machine, relies on the impact of the tail gas on the impeller to do work, promotes the operation of the expansion machine, balances the power distribution of the whole shaft of the air compressor unit, reduces steam consumption, and makes full use of the expansion of the high-temperature tail gas to do work and recover energy to achieve the purpose of energy saving and consumption reduction. The rated compression ratio of the air compressor used in this system is 2.56, the air flow rate is 151 g/s , it is required to work in the high efficiency area of the air compressor, the rated point of the whole machine efficiency is 78 %, the energy recovery is 5.71 kW , the total power consumption of the air compressor is 12.78 kW , the recovery energy is high, the overall power consumption is small. Reserve 12 % surge margin, 19 % speed margin, with 1500 m plateau operation capacity. The MAP of air compressor is shown in Fig. 7, and the main index parameters are shown in Table 5.

The anode cycle mode recycles the unreacted hydrogen from the stack to the hydrogen inlet of the stack through the hydrogen cycle system, thereby improving hydrogen utilization and hydrogen safety. The fuel cell engine developed this time uses the hydrogen supply and return assembly of therail + injector + ejector.

After the front hydrogen bottle passes through the two-stage pressure reducing valve, the pressure reaching the hydrogen injectionrail is less than 20 bar, and the pressure of the hydrogen injectionrail can be regulated in the range of $10\text{--}20 \text{ bar}$, so the rail pressure is designed to be 15 bar. According to the hydrogen consumption demand of the stack, the hydrogen consumption of the three stacks at the rated point is 2.48 g/s , that is, the hydrogen consumption of the single stack is 0.83 g/s , and the ejector leaves 20 % margin, that is, the design hydrogen flow of the single ejector is 1 g/s , and the static flow of the three ejector assemblies is 3 g/s . Therefore, according to the ejector primary flow pressure (hydrogen injectionrail pressure), stack pressure and stack hydrogen consumption, the structural size of the ejector is calculated, including the interceptor port area, the cross-sectional area of the equal volume mixing chamber and the axial length of the ejector. Through one-dimensional and three-dimensional simulation and repeated test verification, the final design scheme of the ejector was confirmed, as shown in Fig. 8 below.

The cooling system mainly consists of stack cooling and intercooler cooling. According to the calculation [18], the maximum heat dissipation demand of the stack is 165.65 kW , and the heat dissipation demand of the intercooler is 16.52 kW , so the total heat dissipation demand of the system is about 180 kW , 50 % glycol aqueous solution is used as coolant, and the maximum flow rate is about 425 L/min . Considering the flow resistance of the stack, radiator and three-way valve, the pump head needs to be about 18.47 m [19]. The selected pump's MAP is shown in Fig. 9, and the main index parameters are shown in Table 6.

Humidifiers are evaluated according to the dew point approach temperature method. The approximate temperature of the dew point is equal to the value of the dew point of the wet air inlet minus the dew point of the dry air outlet, which simply expresses the humidifying ability of the humidifier. Considering dry air inlet carries very little water, it can be negligible. Therefore, the dew point approach temperature is how much water carried by wet air (wet dew point) can be transferred to the dry air side (dry dew point). Generally, the difference between the dew point of the humidifier and the dew point required by the stack is within 2°C . This method needs to provide the measured dew point temperature data of the stack.

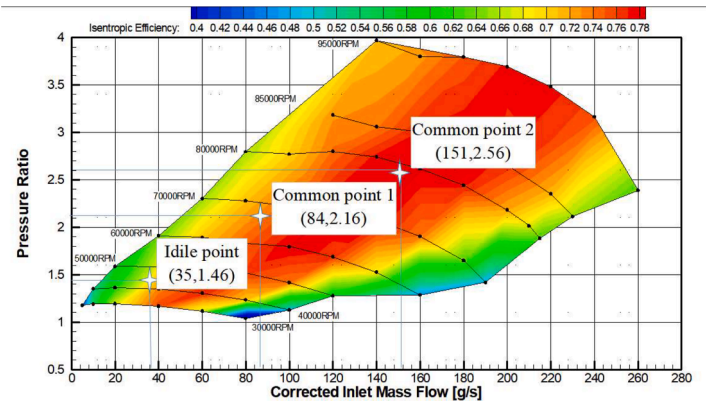


Fig. 7. Map of air compressor.

Table 5
Main index parameters of air compressor.

Name	Parameter	Remark
Air inlet pressure, kPa	70–100	–
Operating ambient temperature, °C	–30–65	–
Rated flow, g/s	161	pressure ratio 3.05
Rated power, kW	18	sea horizon
Air compressor weight, kg	18.5	–
Controller weight, kg	≤4	–
Coolant temperature, °C	–30–65	–
Total coolant flow rate, L/min	6–10	–
Maximum total heat dissipation power, kW	2.2	Full power
Speed, r/min	30,000–90,000	–
Storage temperature, °C	–40–85	–
Operating temperature, °C	–30–65	–
Motor temperature sensor	PT1000	–
Maximum pressure of cooling medium, kPa	250	–

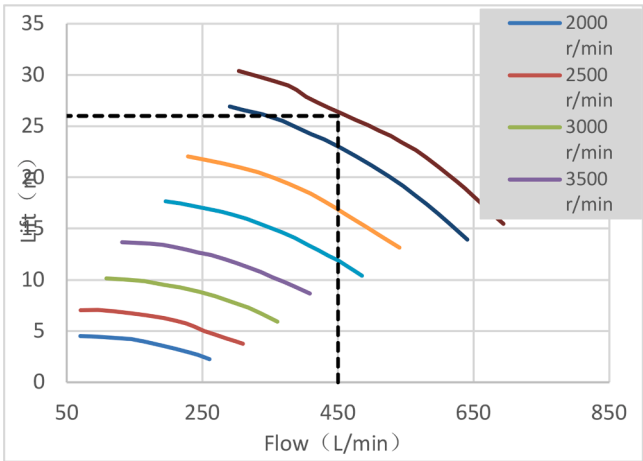


Fig. 9. The location of the design point in the flow - lift map of the pump.

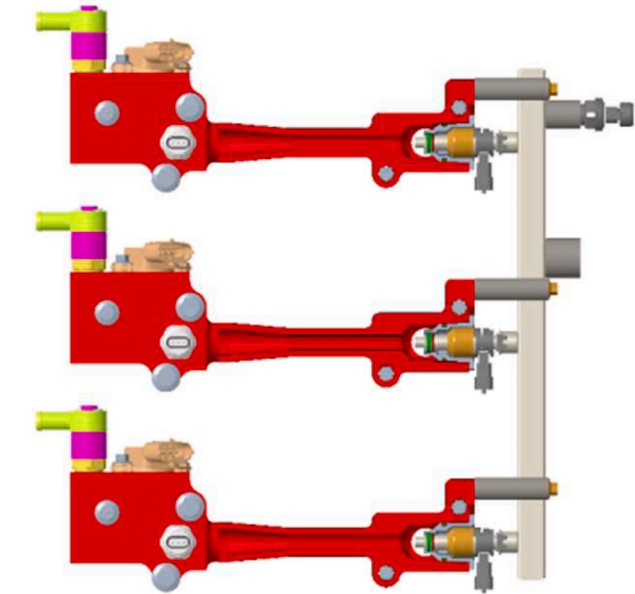


Fig. 8. The final design scheme of the rail + injector + ejector assembly.

In addition to the dew point approach temperature as an indicator for evaluating humidification capacity, humidification efficiency is also a commonly used indicator. The humidification efficiency is the ratio of moisture content of dry air to moisture content of wet air (moisture content ratio). By comparing the performance requirements of the humidifier with the measured performance of the humidifier, a relatively

Table 6
Main parameters of the water pump.

Name	Parameter
Suitable medium	Deionized water or FC antifreeze
Operating temperature	–40–95°C
Rated head	23 m, Peak-32 m
Rated flow	450 L/min
Rated speed	5000 rpm(2000–5300)rpm可调
Supply voltage	450–750 V
Power	Rated-2.9 kW, Rated-3.5 kW.
Communication mode	CAN2.0A/B 250 kbps or 500 kbps
Level of protection	IP67
Weight	12 kg
Noise	≤70 dB
Defensive function	Over pressure, under pressure, over temperature, high pressure anti-reverse connection, no-load, overload

suitable humidifier is selected. Generally speaking, the humidification efficiency of the market product humidifier is much higher than the humidification efficiency of the demand, so the humidification program design is extremely important. In this scheme, the dry-blown solenoid valve is connected in parallel at the humidifier to regulate the overall intake humidity.

Other components are matched according to the main parameters. The key index parameters of other components of multi-stack common rail fuel cell engine are shown in Table 7.

4.3. Physical design

The multi-stack common rail fuel cell engine is based on the existing platform of a commercial vehicle manufacturer and adopts the forward design method to adapt to the vehicle chassis layout space as much as possible. Due to the large number of air subsystem components and large

Table 7

Key indicators of other components for the fuel cell engine.

Name	Key parameters
Hydrogen concentration sensor	Detection range: 0–40,000 ppm. Detection precision: 0–1000 ppm. Starting time: 2 s. Response speed: 3 s.
Voltage monitor(CVM)	Service environment: −35–85°C, 0–95 % RH(%). Operating temperature: −40–85°C. Input voltage: DC6–36 V. Channel for collection: 93.
Air filter	Acquisition voltage range: −4.5–4.5 V. Acquisition voltage accuracy: ±10 mV. Boundary dimension: $\Phi 248 \times 415.3$ mm. Inlet and outlet dimensions: 80 mm. Operating temperature: −40–80 °C. Rated flow: 480 Nm ³ /hr. Initial pressure difference: ≤2.5 @360 Nm ³ /hr. Filter efficiency: >98@0.3 μ m. Working pressure: 20–300 kPa. Operating temperature: −40–105°C.
Stop valve	Relative humidity: 0–100 %.
Solenoid valve	Close response: <100 ms.
Bypass valve	Open response: <150 ms.
Back pressure valve	Maximum torque of valve plate: >3.2 Nm.
Gas-water separator	Separation efficiency: ≥95 %.
Hydrogen water separator	Flow: ≥160 LPM.
Hydrogen pressure relief valve	Working pressure: 0–3.5 bar. Separation efficiency: ≥90 %.
Hydrogen stop valve	Maximum hydrogen flow: 2600 SLPM. Operating gas temperature: −40–100°C. Water storage capacity: ≥285 mL.
Purge valve	Material: Copper.
Drain valve	Operating temperature: −30°C–80°C.
Three-way valve	Initial relief pressure: 280 kPa.
Coolant clarifier	Full open pressure: 330 kPa.
PTC heaters	Flow: ≥350 L/min@Air.
Deionizater	Service life: ≥50,000 times.
	Weight: 0.81 kg.
	Working pressure: 0–16 bar.
	Medium temperature: −30–120°C.
	Starting time: 20–40 m.
	Closing time: 80–100 m.
	Working voltage: 12 VDC/24 VDC.
	Starting/Closing time: 20/22 ms.
	Opening frequency: ≤1 time/s.
	Solenoid valve diameter: 4 mm.
	Heating power: 52 W.
	Maximum temperature resistance: 120°C.
	Heating rate ≥1°C/s.
	Rate voltage: 24 VDC.
	Voltage range: 18–32 VDC.
	Rate current: ≤1 A.
	Internal leakage: ≤300 ml/min @100 kPa.
	Air tightness: no leakage @±300 kPa within 60 s.
	Pressure drop: ≤30 kPa@200 L/min.
	Filter medium: 50 % glycol solution.
	Coolant temperature: −45°C–95°C.
	Rated flow: 220 L/min.
	Original resistance: ≤10 kPa.
	Filtration accuracy: 100 mesh.
	Power: 13.5–16.5 kW.
	Operating voltage: 400–750 V DC.
	Maximum current: ≤27.5 A.
	Coolant temperature: −40–90°C.
	Burst pressure: 5 bar.
	Temperature resistance: −50–93°C.
	Medium dosage: > 750 meq.
	Conductivity: ≤2 us/cm.

interface diameter, the air subsystem scheme is preferred, and the hydrogen subsystem components and interfaces are smaller and integrated on the multi-stack manifold module. The thermal management subsystem is arranged with the design concept of "stitching", and the final outline size meets the boundary requirements of the vehicle, as shown in Fig. 10.

Based on the design concept of traditional internal combustion engine, it innovatively adopts multi-stack common rail injection technology, manifold module integration technology, modular stack technology, air compressor energy recovery technology, humidifier - intercooler integration technology and multi-stack design concept to achieve modular and standardized stack. The technical problems such as uneven gas distribution, difficulty in hydrothermal management, high sealing requirements and poor production adaptability are solved, and the flexibility of the stack application is improved. Three stack manifold module is highly integrated, reduce the number of components, reduce costs, reduce power loss, make the product structure more compact. See Fig. 11.

The hydrogen subsystem adopts the integrated manifold module design of the integrated hydrogen supply system, and innovatively proposes the multi-stack common rail hydrogen injection system to achieve the multi-stack homogeneous and accurate hydrogen supply. At the same time, pulse injector is adopted to improve the hydrogen return flow of the pilot by using pulse energy. The integrated manifold module reduces the sealing surface, ensuring the compactness and tightness of the fuel cell engine. See Fig. 12.

The centrifugal air compressor with energy recovery turbine is adopted in the air subsystem, which reduces the energy consumption of the air compressor and improves the fuel utilization rate. Meanwhile, the self-developed gas-water separator is adopted to improve the durability of the air compressor. Intercooler and humidifier integrated design, compact structure, reduced volume, while reducing parts and sealing surface, improve the reliability and safety of the system, see Fig. 13.

The thermal management subsystem matches the high-voltage controllable electric water pump and the flow control of the intelligent management module to realize the accurate control of the internal temperature of the stack and ensure the consistency and uniformity of the temperature of multiple stacks. The water pump is arranged at the bottom to facilitate water refill and the rubber pad is fixed on the bracket to play the role of insulation and vibration reduction. The exhaust port is designed to connect with the expansion tank at the highest point to facilitate the exhaust of the entire system. See Fig. 14.

Considering the space restrictions such as vehicle ground clearance and cab height, the general three-stack and the following system is a medium power scheme, and the stack is arranged horizontally, so that the vertical distance between the stack and the size of a single stack is small, usually does not affect the uniformity of water and gas distribution, and thus affects the output performance. However, for high-power four-stack or more system schemes, the stack is generally vertically arranged, which is more conducive to the drainage and purging of the fuel cell system, and can slow down the micro-reaction inside the stack after shutdown to extend the life. In addition, in the vertical arrangement scheme, the weight of the stack can be evenly distributed on the end plate to avoid the "collapse" phenomenon caused by the horizontal arrangement of the high-power stack. See Fig. 15 for the stacking scheme.

5. Multi-stack fuel cell engine performance test and analysis

5.1. Test objective

Based on the above three fuel cell engine design scheme, the whole fuel cell engine trial production is completed. Due to the limitation of hydrogen supply and heat dissipation capacity in the laboratory, this test analyzed the performance and consistency performance of multi-stack

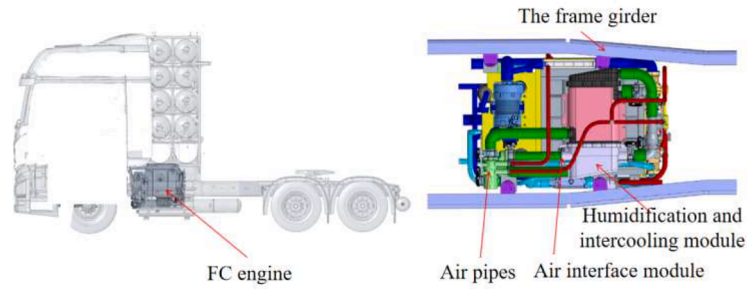


Fig. 10. Multi-stack fuel cell engine carrying vehicle layout.

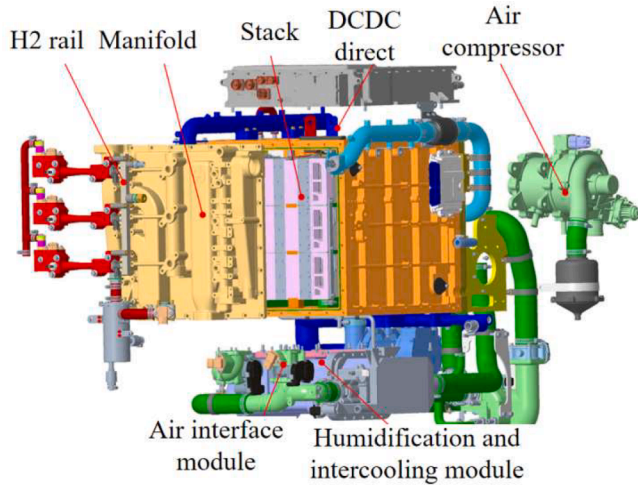


Fig. 11. General layout scheme of multi-stack fuel cell engine.

fuel cell engines through the measured performance data below 80 kW, and verified the advantages of multi-stack fuel cell engines, including basic detection, starting performance, steady-state performance, etc. The operating input conditions are air flow, air inlet pressure, air outlet pressure, air compressor speed, back pressure valve opening, hydrogen inlet pressure, hydrogen outlet pressure, and coolant temperature. The output data are stack output voltage, stack output current, stack power, air compressor power consumption, water pump power consumption, DC/DC power consumption, and hydrogen utilization. Then the system power and efficiency are obtained.

5.2. Test method

The fuel cell engine coolant is filled before the test, and the fuel cell engine can work after passing hydrogen, and then the fuel cell engine is sealed for 12 h, and no changes should be made to the fuel cell engine after the seal. Test items and test methods are shown in Table 8.

5.3. Test data result and analysis

The basic test results are shown in Table 9, which meet the design requirements.

Starting performance test data is shown in Fig. 16, the room

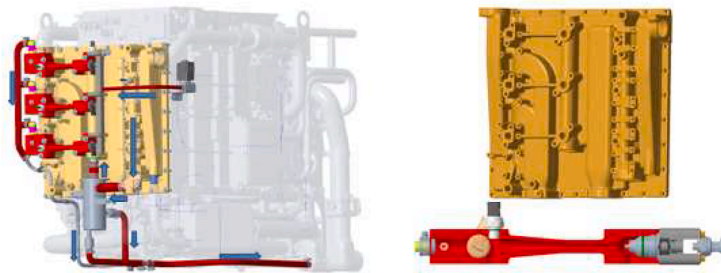


Fig. 12. Design scheme of hydrogen subsystem and front-end manifold module.

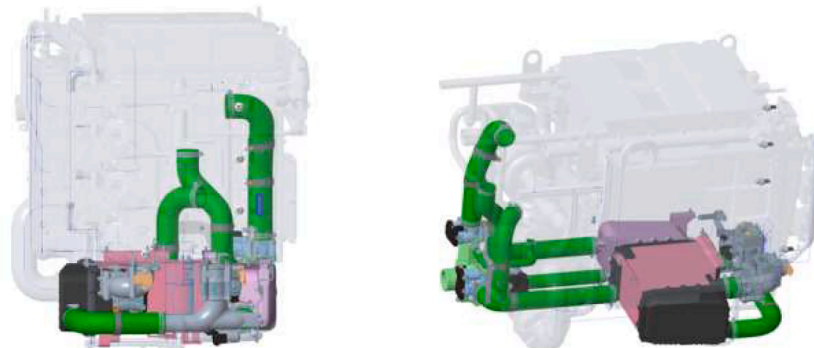


Fig. 13. Design scheme of air subsystem.

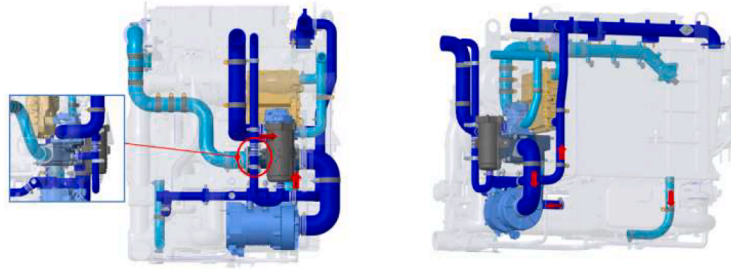
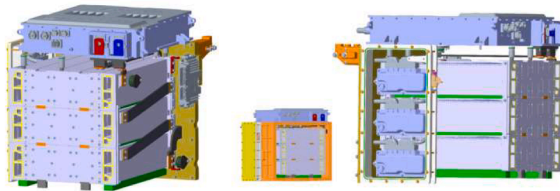
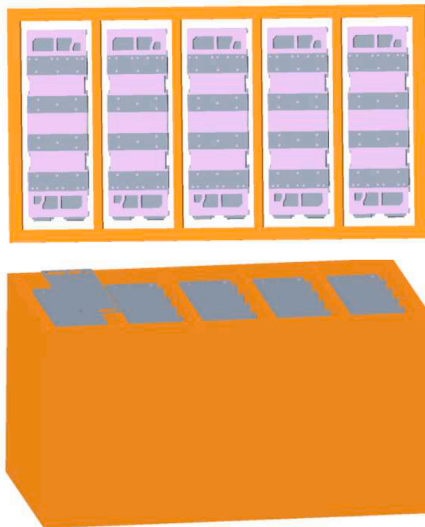


Fig. 14. Thermal management system design scheme.



(a)The three-stack and following system scheme for medium power



(b)The four-stack or more system scheme for high power

Fig. 15. Multi-stack fuel cell engine cabinet stacking arrangement scheme.

temperature start time is 45 s, 0 s start, 45 s room temperature start complete. The output power curve of the system during the room temperature start-up test is shown in the figure. The starting time of the heat engine is 26 s, the machine starts at 0 s, and the 26 s heat start is completed. The output power curve of the system during the heat engine start-up test is shown in the figure. According to the starting process (see Fig. 17) and the comprehensive analysis of starting time, the starting steps of multi-stack fuel cell engines are basically similar to those of single-stack fuel cell engines and the starting performance is comparable [20].

The performance of the steady-state constant-condition system is shown in Fig. 18. The multi-stack fuel cell engine runs for 8 min at 90 A and 8 min at 150 A. From the test data, it can be seen that the multi-stack fuel cell engine runs stably for a long time, and the power of the whole machine changes within $\pm 1\%$, which is equivalent to the stability of the single stack fuel cell engine. The efficiency of the whole machine has obvious advantages over the single stack fuel cell engine of a certain manufacturer at the same current density.

The performance of steady-state lift conditions is shown in Fig. 19. The multi-stack fuel cell engine runs from 40 A to 180 A with good

Table 8

Performance test items and test methods.

Test item	Test method	Data recording
Mass	Use an electronic scale to weigh.	M_e
Gas tightness	Close the engine hydrogen exhaust valve, drainage valve, back pressure valve; Open the anode stop valve, open the cathode inlet stop valve, control the hydrogen injection valve to make the system anode gas (N_2) pressure reach 0.5 bar, and close the hydrogen injection valve, anode stop valve and cathode inlet stop valve after the pressure reaches 10 min.	P_{a-in} P_{c-in}
Insulativity	Adjust the electrical voltage of the insulation resistance instrument to 1000 V; Clamp the positive and negative electrodes of the insulation resistance meter to the positive electrode and the housing of the system respectively; Clamp the positive and negative electrodes of the insulation resistance meter to the negative electrode and the housing of the system respectively.	R_a R_c
Normal starting	The system is in the state of coolant filling, standing at room temperature for 12 h; Send the start command to complete the start without manual intervention; After the fuel cell engine starts, idle for 1 min.	M_{H2} T_{out} I_{stack} U_{stack} P_{sys}
warm start	The engine completes the heat engine, and the coolant temperature reaches $60^\circ C$; After the heat engine is finished, stop the machine. Send the start command to complete the start without manual intervention; After the fuel cell engine starts, idle for 1 min.	
Steady state constant condition test	After the engine heat engine is completed, enter the idle state; Make the system automatically lift to 90 A and run stably for 8 min; Make the system automatically lift to 150 A and run stably for 8 min.	M_{H2} I_{stack} U_{stack} P_{sys}
Steady-state lifting condition test	After the engine heat engine is completed, enter the idle state; Adjust the working conditions of the system to change the power generation state of the stack; Set the 40 A, 60 A, 80 A, 100 A, 120 A, 150 A, and 180 A operating points to the stable state and run for 1 min.	

Table 9

Data results of basic test items.

Test item	Test result
Mass	The fuel cell engine has a net weight of 400 kg.
Gas tightness	After 10 min, the anode gas pressure decreased to 0.35 bar.
Insulativity	Resistance between the positive electrode and the housing: 2.3 M Ω . Resistance between the negative electrode and the housing: 0.4 M Ω .

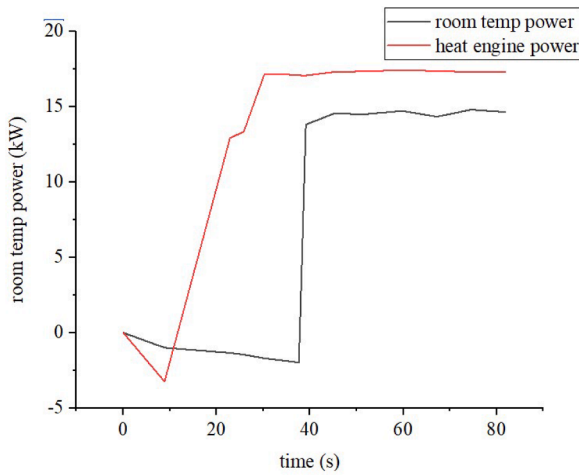


Fig. 16. Fuel cell engine startup performance of the start test.

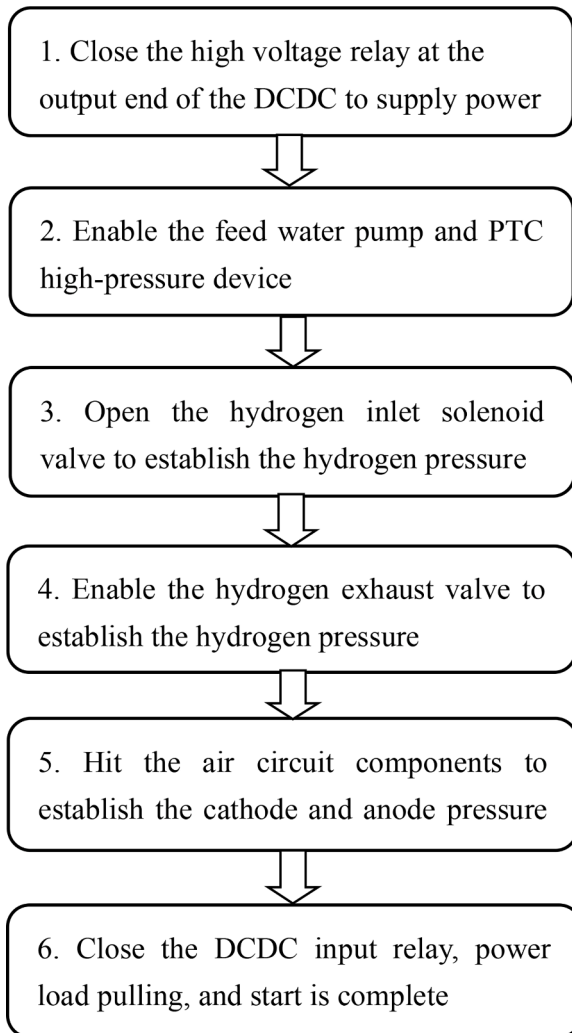


Fig. 17. Start-up process of multi-stack common rail fuel cell engine.

performance. After comparison and analysis with the single stack fuel cell engine of a certain manufacturer, the efficiency of the whole machine is about 2–3 % higher than that of the single stack fuel cell engine at the same current density, as shown in Fig. 20.

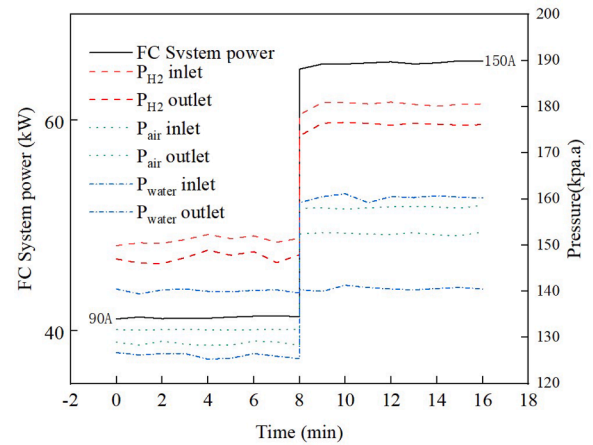


Fig. 18. Steady-state test performance under constant conditions.

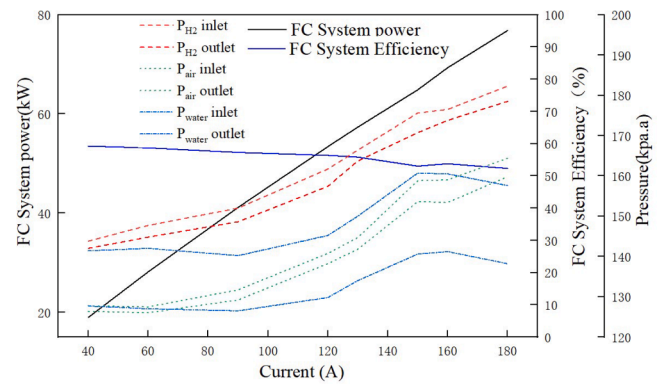


Fig. 19. Power and efficiency of multi-stack common rail fuel cell engines.

5.4. Multi-stack fuel cell consistency evaluation

Since the fuel cell is formed by three stacks in series, each stack contains 210 single cells, a total of 630 cells. Each stack matches a monitor, a total of three inspectors. In order to ensure the volume power density of the fuel cell system, the external accessories should be as small as possible to ensure the basic function, so the number of fuel cell inspection channels is 90. The two sides of the stack are metal end plates, which have large heat dissipation and lower temperature than the central part of the fuel cells. The performance of the single cells near the two sides of the end plate are usually lower than that of the middle part, which has obvious end-plate effect [21]. Therefore, the single cells before and after the end plate needs to be monitored separately. The fuel cell stack voltage monitoring adopts the mode that the first 15 cells each occupy a single channels to achieve a single cell check. The middle 180 cells, each three cells occupy a channel, to achieve three pieces of one inspection, a total of 75 channels. The last 15 cells are the same as the first 15 cells. The monitor defines that the number of a single cell at the inlet and outlet side is number 1, the first 15 cells are numbered from 1 to 15, the middle 180 cells are numbered from 16 to 75, and the last 15 cells are numbered from 76 to 90. In this paper, the voltage consistency of the fuel cell system is judged by graph method, cell voltage fluctuation rate and maximum voltage deviation rate.

Generally, the graph method and cell voltage fluctuation (C_V) can be used as the evaluation criteria for the consistency of fuel cell stack [22]. The advantage of the graph method is that it is concise and explicit, and the uniformity of the fuel cell can be visually seen. The horizontal coordinate is the unit voltage serial number, and the vertical coordinate is the unit voltage, which directly shows the performance changes of each

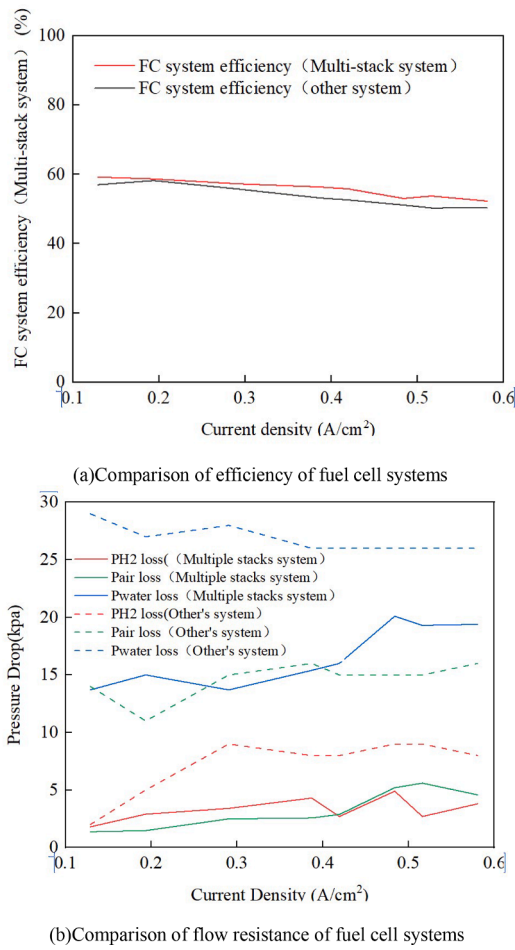


Fig. 20. Comparison of multi-stack system and single-stack system.

cell under different working conditions. The voltage distribution of the three stacks under different current is shown in Fig. 21.

In summary, the following conclusions can be drawn:

- (1) The stack consistency is related to the current of the fuel cell. When the current is low, the voltage fluctuation between each single cell is not obvious. With the increase of the current, the voltage fluctuation of different cells are to increase, and the stack consistency becomes worse. The test results are consistent with those of Professor Hou Yongping [23].
- (2) The single cells performance deviation between the front and rear sides of the stack are large, especially the performance of No. 83, No. 87, No. 88 and No. 89 of the second stack decreases more significantly with the increase of current. There are three possible reasons, first, it may be because of the low concentration of oxygen and hydrogen at the end plate of the stack, and the insufficient gas supply leads to the decrease of reaction rate. Second, the stack produces a lot of water under a large current, the end of the drainage is difficult, resulting in the electrode reactive center is covered, the reaction rate is reduced; Third, the stack activation is not sufficient before the test, especially the single cells activation of the end plate of the second stack are incomplete, Those catalysts fail to participate in the electrochemical reaction.
- (3) Single stack internal consistency judgment. Above stack 1: The difference between the average value and the minimum value is 6–13 mV, the difference between the maximum value and the average value is 6–14 mV, and the difference between the maximum and the minimum value is 12–27 mV; Above stack 2: The difference between the average value and the minimum

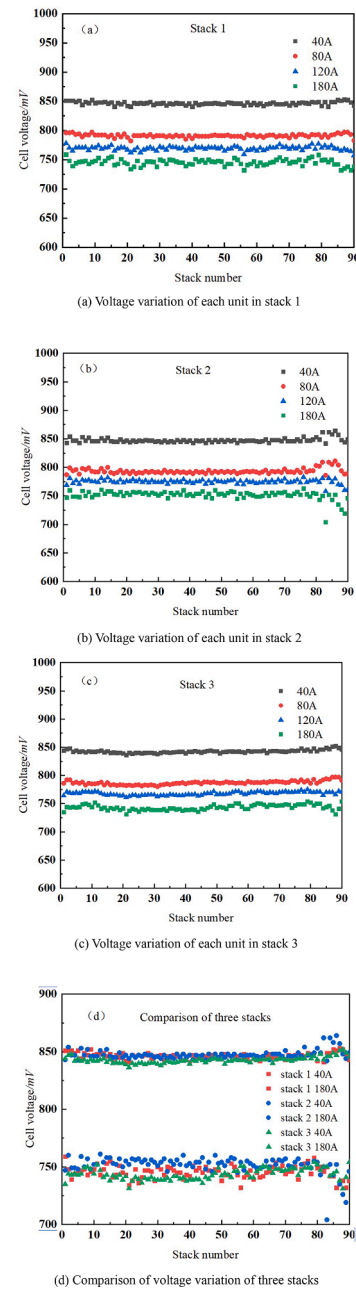


Fig. 21. The voltage distribution of the three stacks under different current.

value is 6–48 mV, the difference between the maximum value and the average value is 8–16 mV, and the difference between the maximum and the minimum value is 22–59 mV; Above stack 3: The difference between the average value and the minimum value is 6–12 mV, the difference between the maximum value and the average value is 7–11 mV, and the difference between the maximum and the minimum value is 14–23 mV. The difference between the No. 1 and No. 3 stacks is basically the same, the No. 2 stack is relatively poor, the No. 2 stack difference > the No. 1 stack difference > the No. 3 stack difference. Remove No. 83, 87, 88 and 89 of the stack 2 (with large fluctuations), the difference between the average value and the minimum value of stack 2 is 5–14 mV, the difference between the maximum value and the average value of stack 2 is 8–18 mV, and the difference between the maximum and the minimum value of stack 2 is 20–25 mV. From the above data, the uniformity of each stack is good.

- (4) Consistency judgment between stacks. The average voltage difference of the three stacks under the current of 40–180 A is not more than 10 mV. If we don't take into account the No. 83, 87, 88, 89 of stack 2, the difference between the minimum and maximum of the three stacks is not more than 15 mV. Therefore, the consistency of the three stacks is good.

Although the graph method is simple and clear, when the voltage difference of the stack unit is imperceptible, the proportionality of voltage cannot be seen directly. And the graph method has not specific index to compare the advantages and disadvantages of the consistency of different stacks. Cell voltage fluctuation (C_v) is the relative standard deviation of the cell voltage, that mean the ratio of the standard deviation of the cell to the average voltage of the cell. If the voltage fluctuation is small, it means that the average value of the voltage offset is smaller, the voltage distribution is less dispersed, and the consistency of the stack is better. On the contrary, if the voltage fluctuation is large, it indicates that the average value of the voltage offset of a single cell is larger, the voltage distribution is larger, and the consistency of the stack is worse. Formula 1 is the specific formula of voltage fluctuation of a single unit.

$$C_v = \frac{\sqrt{\frac{1}{N} \sum_{i=1}^N (U_i - \bar{U})^2}}{\bar{U}} * 100 \% \quad (1)$$

In formula (1), C_v is the voltage fluctuation rate of a single unit; U_i is the voltage of each single cell; N is the number of individual cells in the fuel cell stack; $\bar{U}$ indicates the average voltage of the stack.

Fig. 22 shows the relationship between the three stack's C_v and the current I . The C_v values of the three stacks show the same trend with the increase of current, and the C_v values increase with the increase of current. The reason may be the uneven distribution of gas supply inside the fuel cell, some of the single catalyst is not fully involved in the reaction, or stack generated more water under the large current, causing part of the single air intake channel to be blocked, or the end plate effect caused by uneven temperature. All of the above reasons may lead to differences in the performance of the single voltage, thus affecting the consistency of the fuel cell [24].

At the maximum current of 180 A, the C_v values of the three stacks reach 0.74 %, 1.07 % and 0.65 % respectively, $C_{v2} > C_{v1} > C_{v3}$. The C_v value of the three stacks is low, the maximum is not more than 1.1 %, which proves that the uniformity of each stack is good.

When the four single cells with abnormal performance are removed from stack 2, the C_v values of the three stacks are close to each current point, indicating that the C_v values of stack 2 are larger than those of the

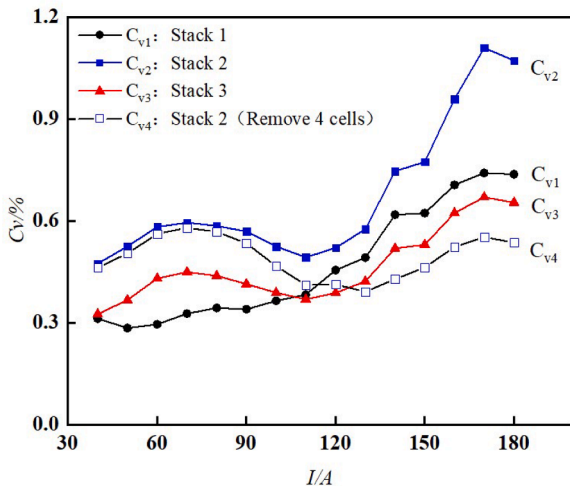


Fig. 22. Relation between voltage fluctuation and current.

other two stacks because the performance of the four cells with abnormal performance is significantly different from the average value.

The maximum voltage deviation rate determines the voltage consistency of fuel cell. According to the research of Hou Yongping on fuel cell stack voltage consistency under vehicle cycle conditions [25], it is found that there are certain defects in the graphical method and the method of unit voltage fluctuation to evaluate the homogeneity of the stack. Therefore, an improved method for consistency evaluation is proposed: the stack life of the fuel cell depends on the single cell with the worst performance, and the degree of deviation from the average value of the minimum voltage of the unit needs to be evaluated. Therefore, the relative difference between the average and minimum voltage of a single cell is defined as the maximum voltage deviation rate (D_v). The larger the D_v value is, it indicates that there is a single cell with poor performance in the stack, and the consistency of the stack is poor. When the D_v value is greater than 1, it indicates that there is a single cell reverse pole, which is easy to cause the stack to burn through. The specific formula is shown in Eq. (2).

$$D_v = \frac{\bar{U} - U_{min}}{\bar{U}} * 100 \% \quad (2)$$

In formula (2), D_v is the maximum deviation rate of voltage; U_{min} indicates the minimum voltage of a single cell.

Fig. 23 shows the relationship between D_v and current I in three stacks. The D_v values of the three stacks show the same trend with the current, and the D_v values increase with the increase of the current. The D_v value of the stack 1 and the stack 3 is basically the same at each current, and the D_v value of the stack 2 is relatively high, mainly because the 4 single cells mentioned above are far from the average value of the stack and meet the definition of the maximum voltage deviation rate D_v . After removing the 4 single cells with abnormal performance, the D_v value of the three stacks is basically the same.

At the maximum current of 180 A, the D_v values of the three stacks are 1.8 %, 6.3 % and 1.7 % respectively, $D_{v2} > D_{v1} > D_{v3}$. The D_v values of the three stacks are low, the highest is not more than 6.5 %, which proves that the uniformity between each single cell of each stack is good.

Based on the above homogeneity evaluation methods, the difference of stack 2 > stack 1 > stack 3 (the difference between the maximum value, minimum value and average value), $C_{v2} > C_{v1} > C_{v3}$, $D_{v2} > D_{v1} > D_{v3}$. The evaluation results of the three kinds of homogeneity are consistent, the homogeneity of stack 3 > stack 1 > stack 2.

Excluding the 4 abnormal cells in stack 2, from the numerical point of view, the difference C_v value and D_v of the three stacks are small, which proves that the consistency between the inner cells of a single stack is good. The difference C_v value and D_v of the three stacks are close,

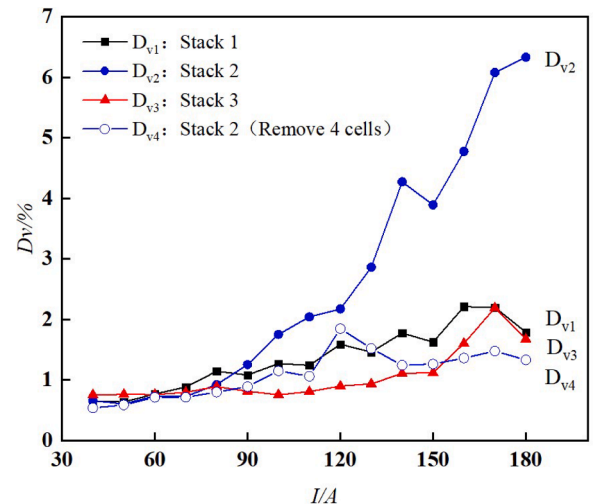


Fig. 23. Relationship between maximum voltage deviation rate and current.

which indicates that the consistency between the stacks is good. The consistency of multi-stack common-rail fuel cells is basically the same as that of ordinary single-stack, and the design of multi-stack common-rail does not affect the uniformity of multi-stack fuel cells.

6. Discussion

- (1) First, from the perspective of system principle and scheme, the circuit of air subsystem, hydrogen subsystem and thermal management subsystem of multi-stack fuel cell engine adopts a parallel scheme. Compared with the single-stack system, the flow resistance of water, hydrogen and air is reduced by about three times. From the perspective of accessory consumption, it will reduce the parasitic power consumption of air compressor and water pump, especially equipped with energy recovery type air compressor. Greatly improve system efficiency.
- (2) Secondly, from the perspective of the overall structure arrangement, the single-stack high-power fuel cell engine is generally horizontal because of the large number of stacked cells and the large length size, and the stack interface is toward the rear of the vehicle, which is not conducive to the stack drainage, while the multi-stack high-power fuel cell engine can achieve a significant reduction in the number of single stack cells and a significant reduction in the length size. The vertical arrangement of the interface facing down can meet the requirements of the vehicle ground clearance and cab height space, and the vertical arrangement is more conducive to the stack drainage, which can greatly shorten the purge time of startup and shutdown, especially improve the cold start performance. At the same time, it can avoid the micro-reaction caused by high humidity inside the stack after shutdown, which is conducive to extending the life of the fuel cell. From the perspective of fuel cell vibration reliability, the weight of all stacked sheets in the vertical layout scheme can be evenly distributed on the end plates, and the load of the end plates can easily ensure the vibration reliability requirements of the fuel cell engine. In the single-stack horizontal layout scheme, the weight of all stacked sheets can only be guaranteed by the clamping force of the stack, and the lateral friction can hardly meet the requirements of the vehicle vibration excitation. It is easy to have the risk of "collapse" of the stack, especially when the length and size of the high-power stack are large.
- (3) Then, the stack adopts a modular design scheme. Through the forward decomposition design of the width of the vehicle girder and the height of the ground and the cab, the membrane electrode and bipolar plate sizes are standardized according to the cutting scheme of the finished materials of the proton exchange membrane, with the goal of reducing waste and maximizing the material utilization rate, so as to realize the standardization and modularization of the entire stack. In the future, the cost of the stack will be greatly reduced after mass, and it will be more suitable for commercialization.
- (4) In addition, after the stack modularization, it is easier to realize the development of high-power fuel cell engines through the series parallel combination of different stacks. There are two main reasons: If a single stack achieves 300 kW, it will require the increase of the single active area and the increase of the number of the whole stack, which will lead to uneven gas distribution in the unit area, local undergas and the risk of partial or single voltage reduction; Multi-stack scheme can easily achieve voltage regulation through different series and parallel combinations, the same power level, the higher the boost ratio, the lower the output current, the smaller the electronic device loss, the voltage ADAPTS to the vehicle platform, saving power consumption.
- (5) Finally, through the test data analysis of multi-stack common-rail fuel cell engine, it can be verified that the multi-stack common-rail system scheme is comparable to that of single-stack fuel cell

engine in terms of start-up performance and power stability. It has obvious advantages in the power and efficiency of the whole machine. The voltage consistency of different cells of a single stack and the consistency between stacks are good.

7. Conclusion

In this paper, a multi-stack common-rail fuel cell system scheme was innovatively proposed to meet the requirements of a manufacturer's 49-ton heavy truck, and a multi-stack common-rail fuel cell engine was developed through forward design and calculation, and the actual test of the pilot prototype was completed. The advantages of the multi-stack common-rail fuel cell system scheme were verified by comparing the measured data with the single-stack fuel cell engine. Including efficiency, consistency, mass and other advantages. The solution is proposed for the future low-carbon application of high-power long-distance heavy trucks.

However, due to the limitation of laboratory hydrogen supply capacity and heat dissipation capacity, the peak performance of multi-stack common rail fuel cell engine has not been fully tested, so further testing and verification are needed.

CRediT authorship contribution statement

Ji Pu: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Qianya Xie:** Conceptualization. **Jun Li:** Writing – review & editing, Resources, Conceptualization. **Ziliang Zhao:** Writing – original draft, Conceptualization. **Junming Lai:** Writing – review & editing, Writing – original draft. **Kang Li:** Writing – review & editing, Project administration, Conceptualization. **Fojin Zhou:** Writing – original draft, Software, Conceptualization.

Declaration of competing interest

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and publication of this article.

Data availability

Data will be made available on request.

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